MTE 203 - Lecture 4

Midterm Exam Spring 2025, Problem 2

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a. Find an equation of the plane through the line of intersection of the planes

$$x - z = 1$$

$$y + 2z = 3$$

and perpendicular to the plane $x + y - 2z = 1$.

b. What is the distance between point $P(2,1,4)$ and this plane.

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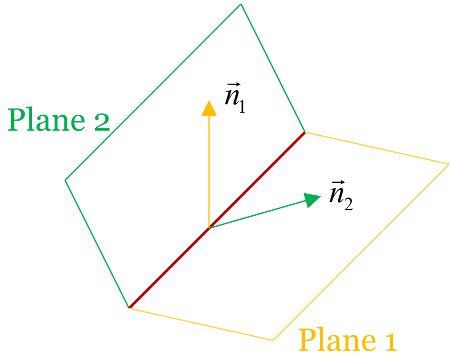
a. From Lecture 3, we learned that to determine the equation of a plane using vectors, we need 2 known vectors in that plane.

First known vector:



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Second known vector:

The normal to the plane we are looking for will therefore be given by,

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b. What is the distance between point $P(2,1,4)$ and this plane.

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Thank You